

KANAK TANWAR

📧 kanaktanwar.tech | 🌐 github.com/kanakOSo1

☎ +91-8860310807 | ✉ kanaktanwarpro@gmail.com | in [linkedin.com/in/kanak-tanwar](https://www.linkedin.com/in/kanak-tanwar)

Experience

Software Development Engineer I | Vetic, Gurgaon, India

July 2026 – Present

- Developing and maintaining **pyvetic**, a shared **backend platform library** providing authentication, database, queuing, observability, and common infrastructure for Vetic's backend applications.
- Engineered **centralized authentication** in **pyvetic** and migrated **15+ backend services** to it, standardizing access control and eliminating duplicated implementations.

Software Development Intern | Vetic, Gurgaon, India

April 2025 – June 2026

- Built a **config-driven DSL rule engine** for appointment exemptions, using declarative rule files supporting multiple operators to enforce outcomes, reducing revenue leakage to **0%** while enabling non-code rule updates.
- Architected an automated **clinic day-closing and revenue reconciliation system**, replacing manual workflows and reducing processing time from **2 hours to 5 minutes** while enabling systematic detection of revenue leakages.
- Scaled **payment processing pipelines** by integrating Pine Labs and Razorpay, handling **80K+ monthly transactions**, improving payment adherence from **90% to 99.7%** and reducing billing mismatches by **40%**.
- Designed a **centralized cash management ledger service** for multi-clinic operations, ensuring consistent tracking of daily cash flows and reducing reconciliation discrepancies across locations.
- Developed an **LLM-assisted OCR pipeline** for extracting structured data from pet vaccination records, reducing manual data entry effort by **85%** and improving data accuracy.
- Implemented **Redis-based distributed locking**, **MongoDB transactions**, **rate limiting**, **multi-layer caching (local + Redis)**, and **asynchronous processing (RabbitMQ, Cron jobs)** to improve backend reliability, correctness under concurrent workloads, and API performance.

AI Intern | Mentorpal.ai

June 2024 – July 2024

- Developed a Chrome extension generating personalized LinkedIn outreach messages using LLMs.
- Built internal resume QA tool using LangChain and LLMs, boosting QA efficiency by 40%.

Projects

Pair Programmer (pp)

[GitHub](#)

- Built an **agentic CLI coding assistant** with a rich TUI, enabling interactive codebase exploration, reasoning, and autonomous code modification using LLMs.
- Designed a **tool-augmented agent architecture** supporting **subagents**, **MCP servers**, and extensible tool integrations for structured multi-step coding workflows.
- Implemented **persistent sessions**, **hooks**, and a **provider-agnostic LLM interface** with streaming responses and dynamic model selection, enabling extensibility and long-running interactive coding sessions.

Git Chat

[GitHub](#)

- Engineered a **RAG-based system** enabling conversational question answering over public GitHub repositories using LLMs.
- Designed a **multi-model backend architecture** with FastAPI, integrating **Weaviate (vector search)**, **MongoDB (metadata)**, and **LangChain (retrieval orchestration)**.

Skills

- Languages & Frameworks** - Python, Go, C++, FastAPI, Django, Flask, Gin
- Databases** - PostgreSQL, MySQL, MongoDB, Redis, Elasticsearch
- Infrastructure & DevOps** - Docker, AWS, Linux, Git, Grafana, Loki, Prometheus
- Misc** - REST APIs, RabbitMQ, LLMs, RAG, MCP, Agentic AI

Education

Degree/Certificate	Institute/Board	CGPA/Percentage	Graduation
B.Tech - CSE	Dronacharya College Of Engineering	9 CGPA	June 2026
Senior Secondary	Manav Rachna International School	96.0%	May 2022
Secondary	Manav Rachna International School	97.6%	May 2020

Achievements

- ICPC Asia West Amritapuri 2023** Regionalist
- LeetCode: **1968** (Knight), CodeChef: **1684** (3★)
- Global Rank **159** in CodeChef Starters 155
- Qualified GATE 2025 — DA (AIR 702), CS (AIR 3050)